PROJECT DOCUMENTATION

Simple VLAN Configuration

Cisco Packet Tracer | Layer 2 Switching | IPv4 Addressing

Project type	Basic LAN segmentation using VLANs
Network device	1 Cisco 2960-series Layer 2 switch
End devices	6 PCs
VLANs configured	VLAN 10, 20, 30 and 40
Simulation tool	Cisco Packet Tracer

A practical demonstration of logical network segmentation, access-port assignment, switch management interface configuration, and basic Cisco IOS verification.

1. Project Overview

This project demonstrates how to create and configure Virtual Local Area Networks (VLANs) on a Cisco 2960-series Layer 2 switch using Cisco Packet Tracer. A single physical switch is divided into separate logical network segments for Students, Faculty, and Management. A fourth VLAN is created for switch ports that are not intended for normal endpoint use.

VLANs help organize devices into separate broadcast domains. Devices connected to different VLANs are logically separated even when they use the same physical switch.

2. Project Objectives

• Create VLANs 10, 20, 30, and 40 with descriptive names.

• Assign switch access ports to the appropriate VLANs.

• Configure a management Switch Virtual Interface (SVI) on VLAN 30.

• Assign static IPv4 addresses to six end devices.

• Place unused Gigabit Ethernet ports in VLAN 40 and administratively shut them down.

• Verify the VLAN and interface configuration using Cisco IOS show commands.

3. Network Components

Component	Quantity	Purpose
Cisco 2960-series switch (S1)	1	Layer 2 switching and VLAN configuration
Student PCs	2	Endpoints in VLAN 10
Faculty PCs	2	Endpoints in VLAN 20
Management PCs	2	Endpoints in VLAN 30
Cisco Packet Tracer	1	Network simulation environment

4. VLAN Design and Port Allocation

VLAN ID	VLAN Name	Switch ports	IPv4 subnet
10	Students	Fa0/1–Fa0/8	192.168.10.0/24
20	Faculty	Fa0/9–Fa0/16	192.168.20.0/24
30	Management	Fa0/17–Fa0/24	192.168.30.0/24
40	Unused	Gi0/1–Gi0/2	No endpoint subnet assigned

The stated port ranges reserve eight Fast Ethernet ports for each user group. The actual six PCs should be connected to ports that belong to their corresponding VLAN.

5. Switch Configuration

5.1 Basic configuration and VLAN creation

```
enable
configure terminal
hostname S1

vlan 10
 name Students
vlan 20
 name Faculty
vlan 30
```

```
 name Management
vlan 40
 name Unused
end
```

5.2 Verify VLAN creation

```
show vlan brief
```

5.3 Assign access ports

```
configure terminal

interface range fa0/1-8
 switchport mode access
 switchport access vlan 10

interface range fa0/9-16
 switchport mode access
 switchport access vlan 20

interface range fa0/17-24
 switchport mode access
 switchport access vlan 30

interface range gi0/1-2
 switchport mode access
 switchport access vlan 40
 shutdown

end
```

Note: The command is **switchport access vlan 20** for the Faculty range. The abbreviated form **switch access vlan 20** in the supplied notes is not the correct Cisco IOS command.

5.4 Configure the management SVI

```
configure terminal
interface vlan 30
 ip address 192.168.30.254 255.255.255.0
 no shutdown
end
```

The VLAN 30 SVI provides an IP address for managing the switch. For the SVI to reach an operational up/up state, VLAN 30 must exist and at least one VLAN 30 access port must have an active link. Remote management from another IP subnet would also require an appropriate default gateway or Layer 3 routing.

6. End-Device IPv4 Addressing

Device	VLAN	IPv4 address	Subnet mask
Student1	10	192.168.10.10	255.255.255.0
Student2	10	192.168.10.11	255.255.255.0
Faculty1	20	192.168.20.10	255.255.255.0
Faculty2	20	192.168.20.11	255.255.255.0
Management1	30	192.168.30.10	255.255.255.0
Management2	30	192.168.30.11	255.255.255.0

The Student2 address has been normalized to 192.168.10.11. The supplied value 193.168.10.11 does not belong to the Students subnet 192.168.10.0/24.

No default gateway is specified in the supplied configuration. A gateway is not needed for communication between devices in the same VLAN and subnet, but is required for communication with other IP networks when routing is configured.

7. Verification and Testing

Run the following commands on switch S1 to inspect the configuration:

```
show vlan brief
show interfaces status
show ip interface brief
show running-config
```

Suggested checks:

• Confirm VLAN IDs 10, 20, 30, and 40 appear in the VLAN table.

• Confirm each Fast Ethernet port range is assigned to the intended VLAN.

• Confirm Gi0/1 and Gi0/2 are assigned to VLAN 40 and are administratively down.

• Confirm interface VLAN 30 has address 192.168.30.254.

• From a PC, test connectivity to another PC in the same VLAN using ping.

8. Expected Network Behaviour and Limitations

Devices in the same VLAN and IP subnet should be able to communicate through the Layer 2 switch when cabling, port status, and IP settings are correct. For example, Student1 should be able to ping Student2.

Devices in different VLANs are separated at Layer 2. They will not communicate with one another in this project because no router or Layer 3 switch inter-VLAN routing configuration has been included. Inter-VLAN communication would require an additional routing design.

The topology documentation is based on the configuration details supplied. Successful ping tests should only be reported after they have actually been performed in Packet Tracer.

9. Security and Good Practices

• Keep unused switch ports administratively shut down.

• Assign unused ports to a designated unused VLAN.

• Use descriptive VLAN names to make administration easier.

• Use consistent IP addressing within each VLAN subnet.

• Save and back up the final Packet Tracer project file and switch configuration.

10. Conclusion

This project provides hands-on practice with basic VLAN implementation on a Cisco Layer 2 switch. It demonstrates VLAN creation, access-port assignment, management SVI configuration, IPv4 addressing, unused-port shutdown, and configuration verification. These are foundational skills for network support and entry-level networking roles.

11. Project Files

Recommended files to include in the GitHub repository:

• README.md – project overview and setup summary

• Simple_VLAN_Configuration.pkt – Cisco Packet Tracer simulation file

• Screenshots/ – topology and verification screenshots

• Simple_VLAN_Configuration_Project_Documentation.pdf – this technical report